

Rachana Sudhakar

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CS graduate student who enjoys building reliable software systems, with experience spanning full-stack web development, robotics, and computer vision.

EDUCATION

Northeastern University

Master of Science in Computer Science

Sept. 2024 – Dec. 2026

Boston, MA

DY Patil School of Engineering and Technology

Bachelor of Technology in Computer Science (**Gold Medalist**)

Aug. 2020 – Aug. 2024

Pune, India

TECHNICAL SKILLS

Core CS: Data Structures & Algorithms, Object-Oriented Design, System Design, Distributed Systems

Languages: Python, JavaScript, TypeScript, Java, C++, Go, Rust, R, PHP

Frameworks: React, Node.js, FastAPI, Flask, Express, Next.js, Redux, Vue, Apache Kafka

Backend & Databases: PostgreSQL, MongoDB, MySQL, DynamoDB, REST API, Prisma, NoSQL, Oracle

Tools & Cloud: AWS, Docker, Git, Terraform, Azure, Linux, Gazebo, Jira, Android SDK

EXPERIENCE

Orpheus Ocean

Robotics Software Engineering Co-op

Jan. 2026 – Aug. 2026

Plymouth, MA

- Engineered real-time telemetry and control systems for autonomous underwater vehicles (**ROS2, Python, FastAPI, React.js, TypeScript**), cutting monitoring latency **98% (5s → 100ms)** for live vehicle health tracking.
- Owned full-stack development of the operator control interface: thruster control, power management, **WebSocket** telemetry, and role-based access, deployed to **13 operators** in production.
- Architected distributed publish/subscribe communication pipelines in **ROS2**, linking frontend applications to autonomous robotic systems for reliable, low-latency service-to-service messaging across the control stack.
- Validated mission-critical workflows across **40+ Gazebo** simulation scenarios and **35+ live AUV deployments** on **NVIDIA Jetson** (tank and boat missions, 2–3 hrs each), surfacing integration risks before live deployment.
- Set up and administered on-prem infrastructure (Intel NUC, LattePanda) hosting **3 internal websites** via **Docker**, and built a NAS-backed storage network for team-wide access.

Tata Consultancy Services

Full-Stack Developer Intern

Nov. 2023 – Apr. 2024

Mumbai, India

- Increased user engagement by **35%** for **2,000+ daily users** by designing UI/UX wireframes in **Figma** and translating prototypes into accessible interfaces using **React.js** across desktop, tablet, and mobile platforms.
- Delivered a real-time IPL Score Tracker serving **5,000+ concurrent users** using **React.js, JavaScript**, and **MySQL**, coordinating with a 3-person database team to achieve **99.9% accuracy** on run rate calculations.
- Processed **10,000+ records daily** for 3 enterprise insurance clients via a BFSI SAP Uploader (**Spring Boot, Maven**), with validation rules to catch inconsistent records.

PROJECTS

Sentiment Aura - AI Emotion Visualizer | React, FastAPI, Python

- Achieved **<500ms** sentiment analysis latency with a real-time emotion app (React, FastAPI, Deepgram speech-to-text, WebSocket), integrating Groq LLM (Llama 3.3 70B) for sentiment detection.
- Developed dynamic Perlin noise visualization using p5.js with **2000 particles** responding in real-time at **60 FPS**, implementing keyword extraction and Web Audio API for microphone capture.

FoodSocial | React, Node.js, PostgreSQL, Prisma, Kotlin

- Built a full-stack SaaS application using React, Node.js, Prisma ORM, and PostgreSQL with Geoapify Places API for restaurant discovery, extending to a mobile client using Kotlin and Android SDK.
- Implemented RESTful APIs with a microservices architecture, deploying via **AWS**, CI/CD pipeline, and Docker containerization, using agile sprints and Jira to track responsive-design work.

Food Calorie Prediction System | TensorFlow, EfficientNetV2B0, Computer Vision

- Built a computer vision system classifying food images into **101 categories** using EfficientNetV2B0 with transfer learning on the Food-101 dataset, achieving **76.70% top-1** and **94.01% top-5 accuracy**.
- Applied data augmentation and fine-tuning to improve generalization, and integrated the USDA FoodData Central API to return nutrition estimates alongside model confidence scores, with the trained model serialized and reused for fast inference.